Komalla Hamsini

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.18
SRI CHAITANYA JUNIOR KALASALA	Class XII (telangana state board of intermediate education)	2024	98.50%
SRI CHAITANYA TECHNO HS BALAJI NGR, MEDCHAL DISTRICT	Class X (telangana state board of secondary education)	2022	10.00%

SKILLS AND TOOLS

- **CS Fundamentals:** Data Structures and Algorithms, Object-Oriented Programming, SQL, Python, C
- **Business and Visualization:** Power BI, MS Excel, MS PowerPoint
- **Libraries & Tools:** Pandas, NumPy, Matplotlib, Jupyter Notebook, GitHub
- **Core Strengths:** Analytical problem-solving, structured thinking, data-driven decision making, cross-functional collaboration
- **Languages:** Python, C, SQL

PROJECTS

IPL Playing XI Recommendation Dashboard | Self Project

- Built an **interactive Power BI dashboard** to recommend the **optimal Playing 11** for IPL franchises by analyzing **280,000+ ball-by-ball deliveries** spanning **2008-2026 seasons**.
- Engineered key **performance metrics** (batting average, strike rate, economy rate, powerplay/death-over impact, and venue-wise form) using **DAX** to quantify **player consistency** and **matchup strength**.
- Framed the dashboard as a **decision-support tool** for franchise management, turning player metrics into **data-driven selection recommendations** for **squad planning and auction strategy**.

Anime Success Prediction & Analysis | Self Project

- Conducted **end-to-end data analysis** on a **7-table relational dataset** (**10,000+ titles**, **150,000+ character-anime links**) using **Pandas** to merge, clean, and structure data across anime, studio, genre, and voice-actor tables.
- Analyzed **studio performance**, **genre trends over two decades**, and **voice-cast influence** to identify the **key factors driving** an anime's critical score and audience popularity.
- Visualized findings using **Matplotlib**, including a **"hidden gems" analysis** highlighting highly-rated but under-watched titles based on **score-to-popularity mismatch**.

Mechanism Simulation Website - Analytical Solver Lead | Faculty-Guided Project

- Owning the **analytical solver module** within a **5-person team** building a **browser-based mechanism visualization tool**, responsible for generating **initial-guess estimates** that feed the system's **real-time kinematic calculations**.
- Applying **structured problem decomposition** to translate **complex nonlinear equations** into a reliable **starting-point algorithm**, directly determining **convergence speed and accuracy** for the tool's live motion outputs.
- **Collaborating with teammates** across simulation modules to **define data interfaces**, **track milestones**, and keep **integration on schedule**.

RELEVANT COURSEWORK

Computer Science: Introduction to Programming

Online Courses: Python, Data Structures and Algorithms, Object Oriented Programming, SQL

Mathematics: Statistics, Probability, Linear Algebra

Business & Management: Introduction to Entrepreneurship, Psychology of Interpersonal Relationships

Mechanical Engineering: Kinematics and Dynamics of Machinery, Manufacturing Science

CERTIFICATIONS

Python Online Certification by AI For Techies-link

SCHOLASTIC ACHIEVEMENTS

- Secured **Gold medal** in Science Talent Search Olympiad (2019)
- Secured **6 Silver medals** in **Talent Search Olympiads (INTSO)**: Science (2018, 2021), Math (2019, 2021), and Aptitude (2019, 2021).

EXTRACURRICULAR

- Won Silver medal in Badminton in Diesta'25-inter department cultural and sports fest
- Active member of the National Sports Organization (NSO) - Badminton.
- Won Silver medal in Dodgeball in Milan'24-inter hostel cultural and sports fest
- **Interests:** Playing Guitar, Badminton, Swimming, Skating, Drawing and Painting, Piano